

Topics of Unit 3 and Classwork Problems

UNIT – III: Orthogonalization, Eigenvalues and Eigenvectors

43-45	T1 Chapter 3: 3.1	Orthogonal Vectors and Subspaces, Orthogonal Bases
46-48	T1 Chapter 3: 3.2	Cosines and Projections onto Lines
49-52	T1 Chapter 3: 3.3	Projections and Least Squares
53-56	T1 Chapter 3: 3.4	Orthogonalization, The Gram-Schmidt Orthogonalization process
57-59	T3 : Handouts	Introduction to Eigenvalues and Eigenvectors, Properties of Eigenvalues and Eigenvectors
60-61	T1 Chapter 3: 5.2	Cayley-Hamilton theorem (statement only), Diagonalization of a Matrix.
62-63	T2: Project 9,10	

Section 3.1(3 sessions)

- 3A to be discussed,
- Useful fact: If non zero vectors are mutually orthogonal linearly independent (Proof not required)
- Orthogonal Subspaces
- 3B to be discussed
- Fundamental theorem of Orthogonality without proof (the truthfulness of the statement can be shown with an example)
- Orthogonal Complement to be discussed

Problem Set 3.1

1. Find the lengths and the inner product of $x = (1, 4, 0, 2)$ and $y = (2, -2, 1, 3)$.
2. Show that the vectors $v_1 = (1, 2, -1)$, $v_2 = (2, -1, 0)$ are orthogonal.
3. Find a vector in R^3 that is orthogonal to both $(1, 1, 0)$ and $(1, 0, 1)$
- 4.

Find a vector x orthogonal to the row space of A , and a vector y orthogonal to the column space, and a vector z orthogonal to the nullspace:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}.$$

5.

Find a basis for the orthogonal complement of the row space of A :

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}.$$

Split $x = (3, 3, 3)$ into a row space component x_r and a nullspace component x_n .

6.

Let $\mathbf{P}$ be the plane in $\mathbf{R}^3$ with equation $x + 2y - z = 0$. Find a vector perpendicular to $\mathbf{P}$. What matrix has the plane $\mathbf{P}$ as its nullspace, and what matrix has $\mathbf{P}$ as its row space?

Section 3.2 (3 sessions)

- Inner product and cosines
- 3G, Law of Cosines
- Projection Onto a line
- 3I, Schwarz inequality
- Projection Matrix of Rank 1
- Transposes from Inner Product

Problem Set 3.2

1. Find the projection of $b=(1,2,7)$ onto the column space of A spanned by $(1,1,-2)$ and $(1,-1,4)$ split b into $p+q$ with p in $C(A)$ and q in Left Null Space of A .
- 2.

Find the matrix that projects every point in the plane onto the line $x + 2y = 0$.

3.

What multiple of $a = (1, 1, 1)$ is closest to the point $b = (2, 4, 4)$? Find also the point closest to a on the line through b .

4.

What matrix P projects every point in $\mathbf{R}^3$ onto the line of intersection of the planes $x + y + t = 0$ and $x - t = 0$?

Section 3.3 Projections and Least Squares (2 sessions)

- Squared error
- Least squares solution
- Orthogonality of a and e
- Least-squares Problems with several variables (Up to 3L)

- Projection Matrices, property 3N (proof not needed)
- Least-Squares Fitting of Data (Up to Example 2)
- Weighted Least Squares, weighted error

Problem Set 3.3

1.

Solve $Ax = b$ by least squares, and find $p = A\hat{x}$ if

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Verify that the error $b - p$ is perpendicular to the columns of A .

2.

Suppose the values $b_1 = 1$ and $b_2 = 7$ at times $t_1 = 1$ and $t_2 = 2$ are fitted by a line $b = Dt$ through the origin. Solve $D = 1$ and $2D = 7$ by least squares, and sketch the best line.

3.

Find the projection of b onto the column space of A :

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}.$$

Split b into $p + q$, with p in the column space and q perpendicular to that space. Which of the four subspaces contains q ?

4.

The vectors $a_1 = (1, 1, 0)$ and $a_2 = (1, 1, 1)$ span a plane in $\mathbf{R}^3$. Find the projection matrix P onto the plane, and find a nonzero vector b that is projected to zero.

5.

Find the projection matrix P onto the space spanned by $a_1 = (1, 0, 1)$ and $a_2 = (1, 1, -1)$.

6.

Find the weighted least-squares solution $\hat{x}_W$ to $Ax = b$:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad W = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Section 3.4 (3 sessions)

Orthogonal Bases and Gram-Schmidt

- Orthonormal vectors
- Orthogonal Matrices
- Rectangular Matrices with orthonormal Columns
- The Gram-Schmidt Process
- The factorization $A=QR$ (Up to $3U$)

Problem Set 3.4

1.

Project $b = (0, 3, 0)$ onto each of the orthonormal vectors $a_1 = (\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ and $a_2 = (-\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$, and then find its projection p onto the plane of a_1 and a_2 .

2.

From the nonorthogonal a, b, c , find orthonormal vectors q_1, q_2, q_3 :

$$a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

3.

Find a third column so that the matrix

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{14} \\ 1/\sqrt{3} & 2/\sqrt{14} \\ 1/\sqrt{3} & -3/\sqrt{14} \end{bmatrix}$$

is orthogonal. It must be a unit vector that is orthogonal to the other columns; how much freedom does this leave? Verify that the rows automatically become orthonormal at the same time.

4.

Find an orthonormal set q_1, q_2, q_3 for which q_1, q_2 span the column space of

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}.$$

Which fundamental subspace contains q_3 ? What is the least-squares solution of $Ax = b$ if $b = [1 \ 2 \ 7]^T$?

5.

Express the Gram-Schmidt orthogonalization of a_1, a_2 as $A = QR$:

$$a_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

Given n vectors a_i with m components, what are the shapes of A, Q , and R ?

6.

(Recommended) Find orthogonal vectors A, B, C by Gram-Schmidt from a, b, c :

$$a = (1, -1, 0, 0) \quad b = (0, 1, -1, 0) \quad c = (0, 0, 1, -1).$$

A, B, C and a, b, c are bases for the vectors perpendicular to $d = (1, 1, 1, 1)$.

7.

Q: Apply Gram-Schmidt orthogonalization process to find an orthogonal basis and then an orthogonal basis for the subspace U of $\mathbb{R}^4$ spanned by a) $v_1=(1,1,1,1)$, b) $v_2=(1,2,4,5)$, c) $v_3=(1,-3,-4,-2)$

Section 5.1(3 sessions)

• Introduction to Eigenvalues and Eigenvectors

• Why do we need eigenvalues and eigenvectors?

- Many "real world" matrices have thousands of rows and columns, which take up a lot of computing power. If we can reduce (or approximate) those large matrices to a single number, it speeds things up considerably.
- **Eigenvectors** are the directions that remain unchanged during a transformation, even if they get longer or shorter. **Eigenvalues** are the numbers that indicate how much something stretches or shrinks during that transformation. These ideas are important in many areas of math and engineering, including studying how systems stay stable and understanding quantum physics. They're also important in fields like machine learning, where they help simplify complex data by reducing its

dimensions.

- **Eigenvalues and Eigenvectors** are the scalar and vector quantities associated with matrices used for linear transformations. The vector that only changes by a scalar factor after applying a transformation is called an eigenvector, and the scalar value attached to the eigenvector is called the eigenvalue.

Our aim is to **replace** our square matrix **A** with some scalar (number) λ for a particular vector **v** so we'll have:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} = \mathbf{v}_R$$

The scalar λ is called an **eigenvalue**, and **v** is called the corresponding **eigenvector** of **A**.

For your better understanding on this refer: [Eigenvalues and eigenvectors - physical meaning and geometric interpretation applet](#)

Properties of Eigenvalues and Eigenvectors

- Eigenvalues of a square matrix A are the roots of its characteristic equation:

$$\det(A - \lambda I) = 0$$

- The determinant of a matrix is the product of its eigenvalues:

$$\det(A) = \lambda_1 \lambda_2 \dots \lambda_n$$

- The trace of a matrix (sum of diagonal elements) equals the sum of its eigenvalues:

$$\text{Tr}(A) = \sum_{i=1}^n \lambda_i$$

- The product of the eigenvalues of a matrix A is equal to its determinant:

$$\prod_{i=1}^n \lambda_i = \det(A)$$

- If a matrix is raised to a power k , its eigenvalues are also raised to that power:

$$\text{Eigenvalues of } A^k : \lambda_i^k$$

- The eigenvalues of an invertible matrix are the reciprocals of the original eigenvalues:

$$\text{Eigenvalues of } A^{-1} : \frac{1}{\lambda_i}, \quad \text{for } \lambda_i \neq 0$$

- The eigenvalues of a transposed matrix remain the same:

$$\text{Eigenvalues of } A^T = \text{Eigenvalues of } A$$

- The eigenvalues of a diagonal matrix are simply its diagonal elements:

$$A = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \Rightarrow \lambda_i = d_i$$

- The eigenvalues of an upper or lower triangular matrix are its diagonal elements:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix} \Rightarrow \lambda_i = a_{ii}$$

- If zero is an eigenvalue of A , then A is singular (non-invertible):

$$\exists \lambda_i = 0 \Rightarrow \det(A) = 0$$

- If v is an eigenvector corresponding to an eigenvalue λ , then **any nonzero scalar multiple** of v is also an eigenvector.

Mathematically, if $Av = \lambda v$, then for any nonzero scalar c ,

$$A(cv) = c(Av) = c\lambda v = \lambda(cv).$$

i.e. for a given eigenvalue there corresponds infinitely many eigenvectors.

Properties of Eigenvectors

- For a given eigenvector x , there corresponds only one eigenvalue λ .
- Eigenvectors corresponding to distinct eigenvalues of a matrix are linearly independent:

$$Av_i = \lambda_i v_i, \quad Av_j = \lambda_j v_j, \quad \lambda_i \neq \lambda_j \Rightarrow v_i \perp v_j.$$

- If a matrix is diagonalizable, its eigenvectors form a complete basis for the vector space:

$$A = PDP^{-1}, \quad P \text{ contains linearly independent eigenvectors of } A.$$

Cayley Hamilton theorem:

Statement: Every matrix A is a root of its characteristic polynomial.

Q:

Verify the Cayley-Hamilton Theorem for the matrix

$$A = \begin{pmatrix} 1 & 5 \\ 4 & 2 \end{pmatrix}$$

by direct calculation.

Solution:

Solution:

Let

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

The characteristic equation is given by

$$\begin{aligned} |A - \lambda I| &= 0 \\ \begin{vmatrix} 1 - \lambda & 5 \\ 4 & 2 - \lambda \end{vmatrix} &= 0 \end{aligned}$$

Expanding the determinant,

$$\begin{aligned} (1 - \lambda)(2 - \lambda) - 20 &= 0 \\ 2 - \lambda - 2\lambda + \lambda^2 - 20 &= 0 \\ \lambda^2 - 3\lambda - 18 &= 0 \end{aligned}$$

This is the **characteristic polynomial equation**.

Now, computing A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (1 \times 1 + 5 \times 4) & (1 \times 5 + 5 \times 2) \\ (4 \times 1 + 2 \times 4) & (4 \times 5 + 2 \times 2) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 20 & 5 + 10 \\ 4 + 8 & 20 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 15 \\ 12 & 24 \end{bmatrix} \end{aligned}$$

Now, computing $A^2 - 3A - 18I$:

$$\begin{aligned} A^2 - 3A - 18I &= \begin{bmatrix} 21 & 15 \\ 12 & 24 \end{bmatrix} - 3 \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix} - 18 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 15 \\ 12 & 24 \end{bmatrix} - \begin{bmatrix} 3 & 15 \\ 12 & 6 \end{bmatrix} - \begin{bmatrix} 18 & 0 \\ 0 & 18 \end{bmatrix} \\ &= \begin{bmatrix} 21 - 3 - 18 & 15 - 15 - 0 \\ 12 - 12 - 0 & 24 - 6 - 18 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Thus,

$$A^2 - 3A - 18I = 0$$

Question:

Verify the Cayley-Hamilton Theorem for the matrix

$$\begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix}$$

Sol. Let $A = \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix}$
 For characteristic equation, $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & -2 \\ -2 & -1-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(-1-\lambda) - 4 = 0$$

$$-2 + \lambda - 2\lambda + \lambda^2 - 4 = 0$$

$$\boxed{\lambda^2 - \lambda - 6 = 0} \quad \text{--- (i)}$$

Matrix A satisfy eqn(i)

$$A^2 - A - 6I = 0$$

$$\text{where } A^2 = \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 4+4 & -4+2 \\ 4+2 & 4+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix}$$

$$\text{LHS} = A^2 - A - 6I$$

$$= \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} - \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix} - 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} + \begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} -6 & 0 \\ 0 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 8-2-6 & -2+2+0 \\ -2+2+0 & 5+1-6 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= 0 = \text{RHS}$$

Question:

Let $A = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix}$, then
 what is the sum of all entries of A^{100}

Hint: use The Cayley Hamilton Theorem

Question:

Use the Cayley-Hamilton Theorem to compute A^{-1} .

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 9 & 2 & 0 \\ 5 & 0 & 3 \end{pmatrix}.$$

Sol:

Determine the characteristic eqⁿ.

$$|A - \lambda I| = 0$$

$$= \begin{vmatrix} 1-\lambda & 1 & 2 \\ 9 & 2-\lambda & 0 \\ 5 & 0 & 3-\lambda \end{vmatrix}$$

It simplifies to

$$= -\lambda^3 + 6\lambda^2 + 8\lambda - 41$$

Acc. to Cayley-Hamilton theorem

$$-A^3 + 6A^2 + 8A - 41I = 0.$$

$$\text{or } -41I = A^3 - 6A^2 - 8A$$

$$= A(A^2 - 6A - 8I)$$

$$\text{or } A^{-1} = \frac{-1}{41}(A^2 - 6A - 8I)$$

Finding $A^2 - 6A - 8I$ comes out as

$$\begin{bmatrix} 6 & -3 & -4 \\ -27 & -7 & 18 \\ -10 & 5 & -7 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{-1}{41} \begin{bmatrix} 6 & -3 & -4 \\ -27 & -7 & 18 \\ -10 & 5 & -7 \end{bmatrix}$$

Problem set 5.1

Q3(Hard copy)

3. If we shift to $A - 7I$, what are the eigenvalues and eigenvectors and how are they related to those of A ?

$$B = A - 7I = \begin{bmatrix} -6 & -1 \\ 2 & -3 \end{bmatrix}.$$

Q7 (Hard Copy)

2.

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

Check that $\lambda_1 + \lambda_2 + \lambda_3$ equals the trace and $\lambda_1 \lambda_2 \lambda_3$ equals the determinant.

3.

Find the eigenvalues and the eigenvectors of these two matrices:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

$A + I$ has the ____ eigenvectors as A . Its eigenvalues are ____ by 1.

4.

(Review) Find the eigenvalues of A , B , and C :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

Section 5.2 (3 sessions) • Exclude Heisenberg's uncertainty principle.

1.

Factor these two matrices into $A = SAS^{-1}$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}.$$

2.

Which of these matrices cannot be diagonalized?

$$A_1 = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix} \quad A_2 = \begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix} \quad A_3 = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}.$$